

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4303

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COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Case Study

Weightage: 10%

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. <div>
- B. <nav>
- C. <section>
- D. <header>

Snippet 4: XML Syntax

- A. <note><to>User</to><from>Admin</from>
- B. <note><to>User</to><from>Admin</from></note>
- C. <note><to>User</from></note>
- D. <note to="User"></note>

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

*I **T. Nikhil Kumar Reddy / 192372024**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

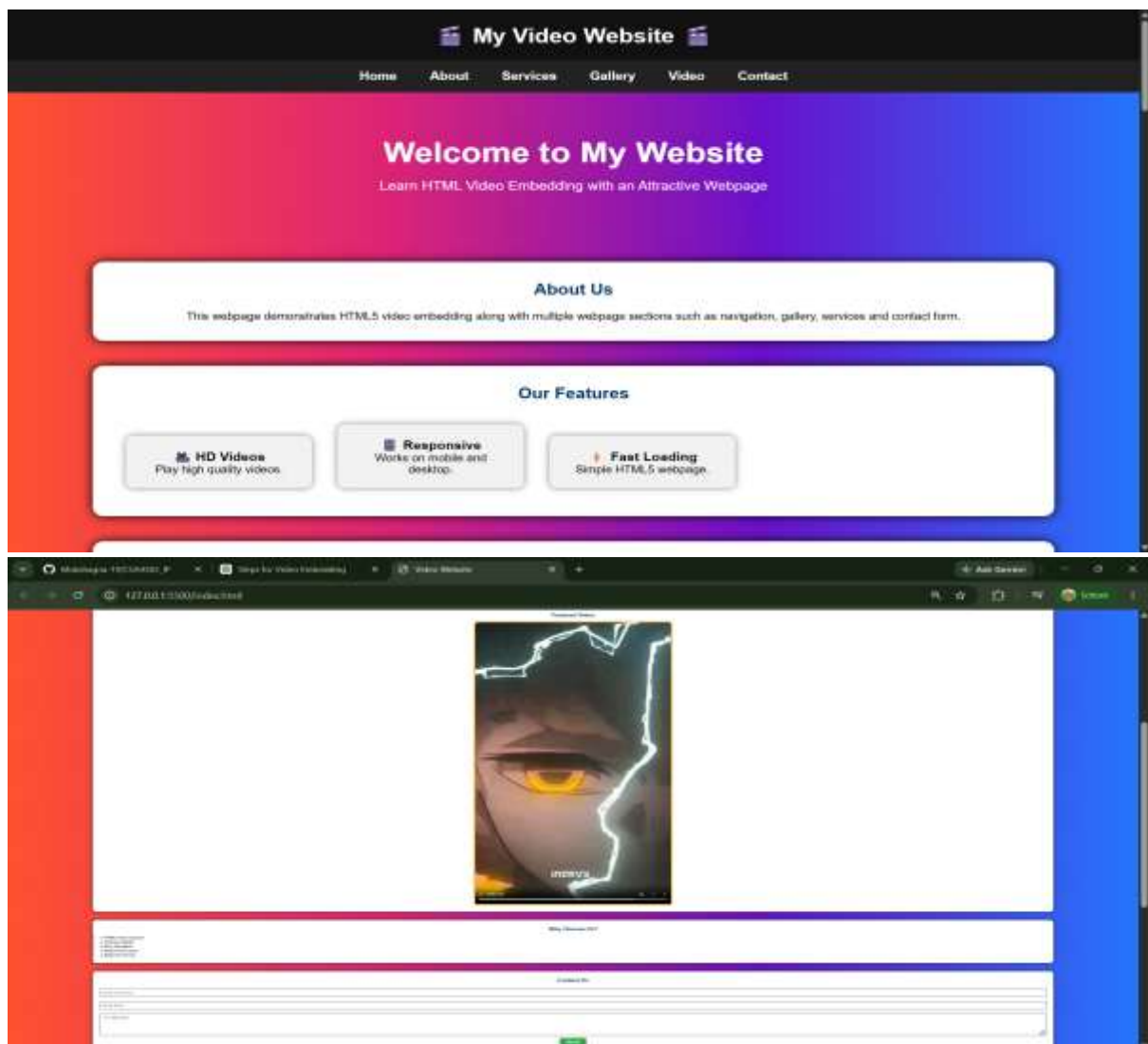
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Nikhil Kumar Reddy

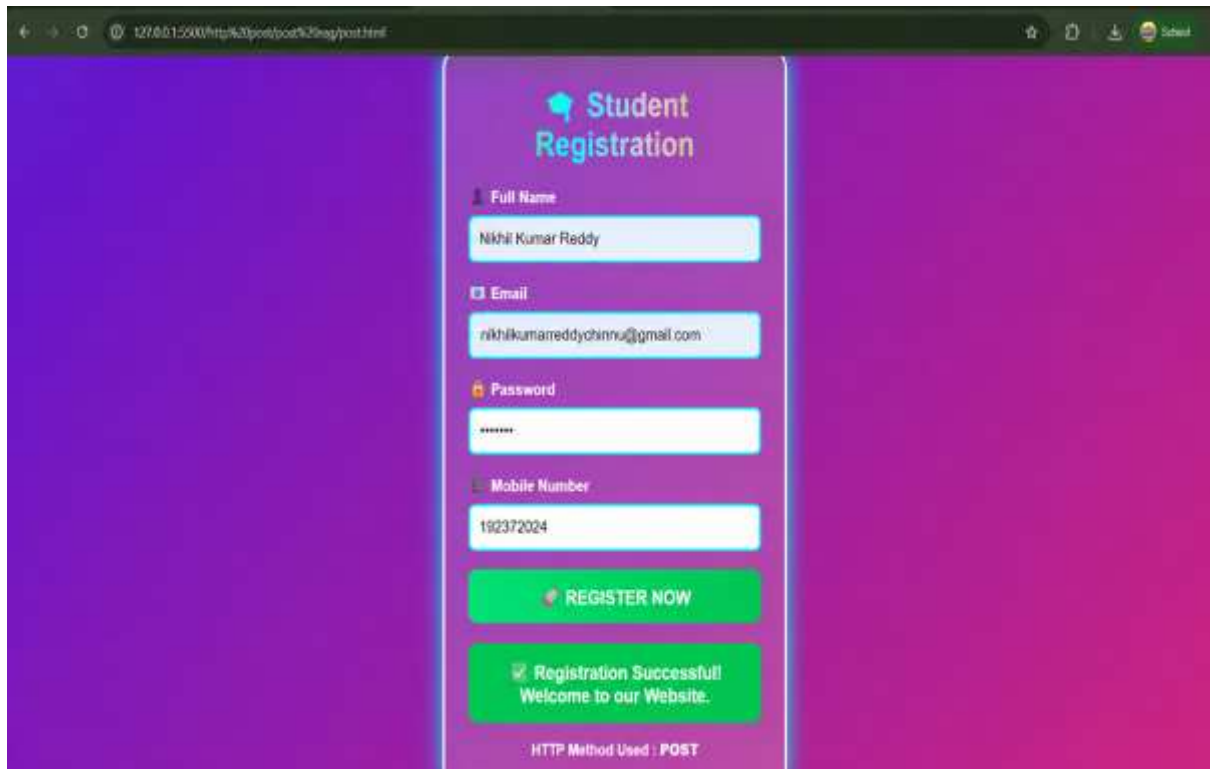
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1.sol:



2.sol:



The screenshot shows a web browser window with the address bar displaying `127.0.0.1:5000/http%3Apost%2Freg%2Fpost.html`. The page has a purple gradient background. In the center, there is a white box titled "Student Registration" with a blue graduation cap icon. Below the title, there are four input fields: "Full Name" (containing "Nikhil Kumar Reddy"), "Email" (containing "nikhilkumareddychinna@gmail.com"), "Password" (containing "*****"), and "Mobile Number" (containing "192372024"). Below these fields is a green button labeled "REGISTER NOW". Below the button is a green box with a checkmark icon and the text "Registration Successful! Welcome to our Website." At the bottom of the white box, it says "HTTP Method Used : POST".

Student Registration

Full Name
Nikhil Kumar Reddy

Email
nikhilkumareddychinna@gmail.com

Password

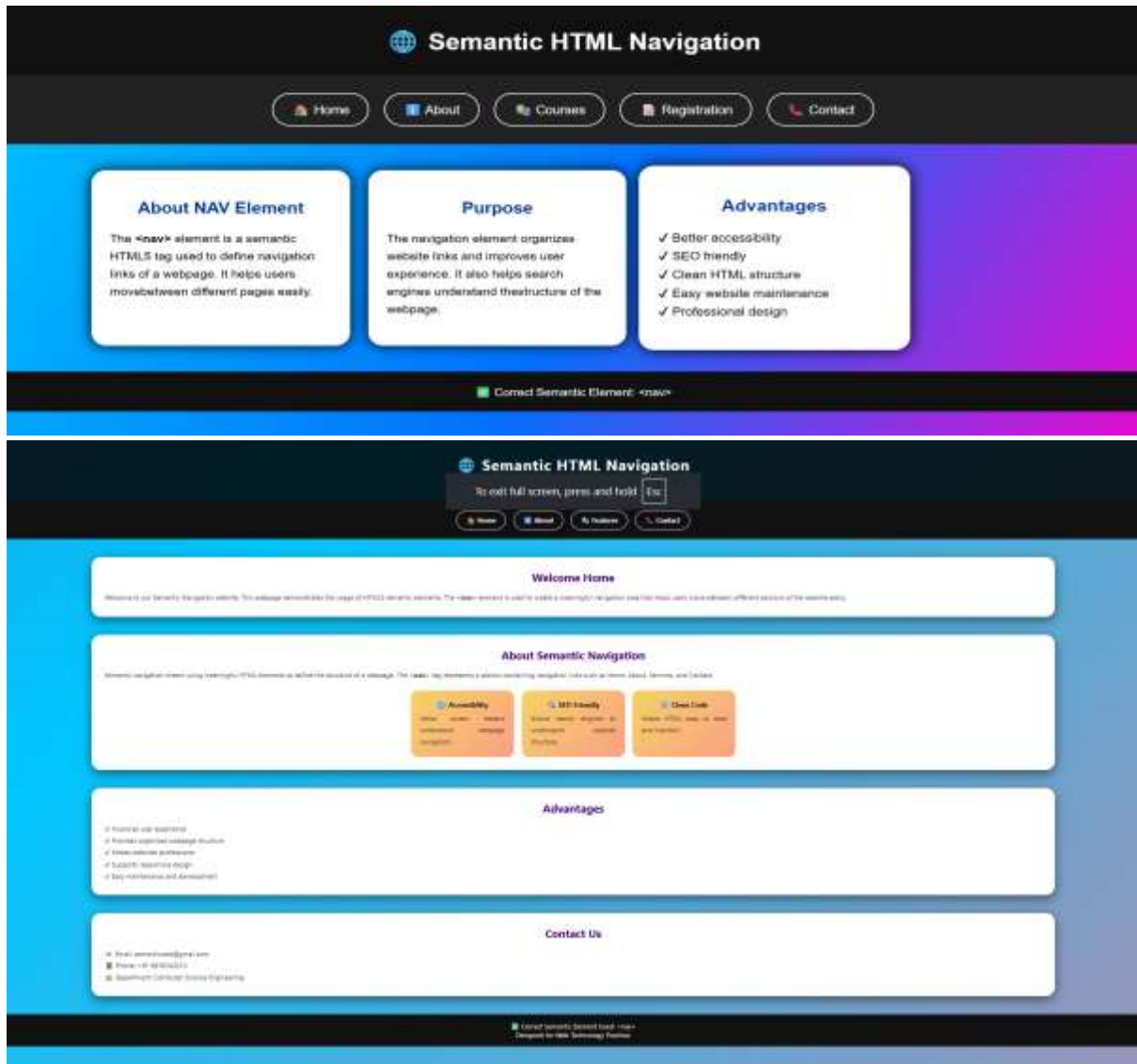
Mobile Number
192372024

REGISTER NOW

☒ Registration Successful!
Welcome to our Website.

HTTP Method Used : POST

3.sol:



4.sol:

Correct Answer: ☒ Option B

<note>

<to>User</to>

<from>Admin</from>

</note>

Explanation

- XML must have one root element.
- Every opening tag must have a closing tag.
- Tags must be properly nested.

VS Code Program (note.xml)

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<note>
```

```
  <to>User</to>
```

```
  <from>Admin</from>
```

```
  <heading>Meeting Reminder</heading>
```

```
  <date>28-07-2026</date>
```

```
  <message>
```

```
    Welcome to XML Configuration File.
```

```
  </message>
```

```
</note>
```

Question 5

Correct Answer: ☒ Option B – HTTP (HyperText Transfer Protocol)

Explanation

HTTP is the standard protocol used for communication between a web browser and a web server. It transfers web pages, images, videos, forms, and other web resources over the Internet.

VS Code Program ([index.html](#))

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>HTTP Web Communication</title>
```

```
<style>
```

```
body{
```

```
margin:0;
```

```
font-family:Arial,sans-serif;
```

```
background:linear-gradient(135deg,#00c6ff,#0072ff,#6a11cb,#ff512f);
```

```
color:white;
```

```
}
```

```
header{
```

```
background:#111;
padding:20px;
text-align:center;
font-size:35px;
font-weight:bold;
}
```

```
section{
width:80%;
margin:30px auto;
background:white;
color:black;
padding:25px;
border-radius:15px;
box-shadow:0 0 20px black;
}
```

```
table{
width:100%;
border-collapse:collapse;
margin-top:20px;
}
```

```
table,th,td{
border:2px solid gray;
}
```

```
th{
background:#2196F3;
color:white;
padding:12px;
}
```

```
td{
padding:12px;
text-align:center;
}
```

```
footer{
background:#111;
color:white;
text-align:center;
padding:15px;
margin-top:20px;
}
```

```
</style>
```

```
</head>
```


<body>

<header>

 HTTP Web Communication Protocol

</header>

<section>

<h2>What is HTTP?</h2>

<p>

HTTP (HyperText Transfer Protocol) is the communication protocol used between a web browser and a web server. It transfers web pages, images, videos, forms, and other web resources.

</p>

<h2>Protocol Comparison</h2>

<table>

<tr>

<th>Protocol</th>

<th>Purpose</th>

<th>Status</th>

</tr>

<tr>

<td>FTP</td>

<td>File Transfer</td>

<td> Incorrect</td>

</tr>

<tr>

<td>HTTP</td>

<td>Web Communication</td>

<td> Correct</td>

</tr>

<tr>

<td>SMTP</td>

<td>Email Transfer</td>

<td>✗ Incorrect</td>

</tr>

<tr>

<td>SNMP</td>

<td>Network Management</td>

<td>✗ Incorrect</td>

</tr>

</table>

<h2>Advantages of HTTP</h2>

Fast communication between browser and server.

Supports GET and POST methods.

Transfers web pages, images, videos, and forms.

Platform independent.

Widely used on the Internet.

</section>

<footer>

Correct Protocol: HTTP (HyperText Transfer Protocol)

</footer>

</body>

</html>